Week 3: Linux Logs, Sysmon, auditd and Snort Setup

Theory Notes, Developing Basic Understanding in SOC

Question 10. Explain the concept of log ingestion and how tools like Wazuh or Splunk collect endpoint and network logs. What is the flow from a log source to a SIEM dashboard?

What is log ingestion

Log ingestion simply means collecting logs from many different places and bringing them into one central system, usually a SIEM. After that, the logs are broken into fields, stored, searched and checked by detection rules, so the analyst can work from one place instead of logging in to every machine.

The flow from a log source to the dashboard

Step	Stage	What happens
1	Log source	Endpoints, servers, firewalls, Snort, Sysmon, auditd and applications create the logs
2	Collection	An agent, forwarder or syslog sender reads the logs
3	Transport	The logs are sent securely to the SIEM server or collector
4	Parsing and normalisation	The raw text is split into fields such as source IP, user and event ID
5	Enrichment	Extra details are added, for example location or threat intelligence
6	Storage and indexing	The logs are stored in a way that makes searching fast
7	Detection and correlation	Rules and analytics look for suspicious patterns and create alerts
8	Dashboard and alerting	Analysts see alerts, charts and search results and begin the investigation

How Wazuh collects logs

- A Wazuh agent is installed on each endpoint. It reads local logs, such as the Sysmon event channel on Windows or auth.log on Linux, and it can also watch for file changes.
- The agent sends the events to the Wazuh manager, by default on port 1514. Agents register with the manager on port 1515.
- The manager decodes the logs, compares them with its rules and creates alerts.
- The alerts are stored in the Wazuh indexer and shown in the Wazuh dashboard.
- Network tools such as Snort can be added by pointing the agent to the Snort alert file.

How Splunk collects logs

- A Universal Forwarder is installed on the endpoint. It reads files or event logs based on `inputs.conf` and sends them to the Splunk indexer, by default on port 9997.
- Splunk can also receive data from syslog, the HTTP Event Collector, and from apps and add-ons.
- The indexer parses the data and stores it in indexes.
- Analysts then use SPL searches, alerts and dashboards on the search head, which has its web interface on port 8000.

Wazuh and Splunk compared

Point	Wazuh	Splunk
Cost	Free and open source	The free version has limits, and full use is paid
Endpoint collector	Wazuh agent	Universal Forwarder
Detection	Built in rules and decoders, with MITRE mapping	SPL searches and correlation rules, with apps for detection content
Storage and search	Wazuh indexer and dashboard	Splunk indexer and search head